



清华大学
Tsinghua University

量子计算研讨课 课程大作业



整体流程

组队：1-2名同学一小组

- 选题：真机实验/论文报告
- 申报选题
 - 通过共享文档上报组队人员与选题
 - 本周五（5.22）完成
 - 做论文报告的同学后续上报论文选择
- 完成作业
 - 第16周课上汇报（6.9）

天衍量子计算云平台真机实验



- 注册平台账号并由助教开通账户权限（已完成）
 - 祖冲之二号，祖冲之三号
 - 中电信天衍量子计算云平台文档：<https://qc.zdxlz.com/learn/#/resource/informationSpace>
- 完成实验大作业中要求的实验（见网络学堂），并逐题回答所有问题
 - 格式不限（LaTeX, Word……）
 - 回答中可以包含图片，表格
 - 确保图表的可读性（不建议直接从云平台网站截图）
 - 将作业中用到的线路通过QCIS格式的代码保存，并在作业的附录中提交
 - 若通过API使用实验平台，请将完整的Python代码/notebook文件打包压缩并在作业附录中提交
- 完成至少三道附加题
 - 完成更多的附加题可以获得额外分数
- 在第16周周五提交到网络学堂
 - 天衍量子计算云平台会不定期维护，请同学们合理安排时间

量子计算论文报告



- 选择1篇论文
 - 旨在对某个主题的研究与理解
 - 不要求完整报告文章的所有内容
 - 最终评分不会考虑论文体量
 - 在共享文档中上报所选论文题目和第一作者姓名
 - How to factor 2048 bit RSA integers with less than a million noisy qubits (Craig Gidney)
 - 如果出现多个小组选择同一篇论文的情况，原则上先到先得，也可以自行协商解决
 - 可以选择未在论文列表中出现的文章
- 撰写书面报告
 - 2500字以内，中英文不限
 - 包括对论文内容的理解，批判性分析，和延伸性思考
- 完成课堂展示
 - 8分钟报告+2分钟提问（可能根据展示小组数量调整时长）
- 在第16周周五将书面报告和课堂展示ppt提交到网络学堂



- Quantum Security and Cryptography
 - Space-Efficient Quantum Algorithm for Elliptic Curve Discrete Logarithms with Resource Estimation (Han Luo)
 - Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (Ryan Babbush)
 - How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits (Craig Gidney)
 - How to factor 2048 bit RSA integers with less than a million noisy qubits (Craig Gidney)
- Quantum Network
 - Quantum cryptography: Public key distribution and coin tossing (Charles H. Bennett)
 - Quantum internet: A vision for the road ahead (Stephanie Wehner)
 - White Paper on Quantum Internet Computer Science Research Challenges (Thomas R. Beauchamp)

论文报告选题示例



- Quantum Algorithms
 - 本方向推荐的参考: <https://quantumalgorithms.org/>
 - Quantum speedup of Monte Carlo methods (Ashley Montanaro)
 - Quantum Perceptron Models (Armando Bellante)
 - Quantum matching pursuit: A quantum algorithm for sparse representations (Armando Bellante)
 - Quantum Recommendation Systems (Iordanis Kerenidis)
 - Quantum SDP-Solvers: Better upper and lower bounds (Joran van Apeldoorn)
- Classical Simulation of Quantum Circuits
 - Qiskit-Aer: <https://qiskit.github.io/qiskit-aer/>
 - Matchgates and classical simulation of quantum circuits (Richard Jozsa)
 - Simulation of low-depth quantum circuits as complex undirected graphical models ([Sergio Boixo](#))



- Gate Decomposition and Circuit Synthesis
 - The Solovay-Kitaev algorithm (Christopher M. Dawson)
 - An Introduction to Cartan's KAK Decomposition for QC Programmers (Robert R. Tucci)
 - Fast and efficient exact synthesis of single qubit unitaries generated by Clifford and T gates (Vadym Kliuchnikov)
- Quantum Information Processing
 - A simple proof of monogamy of entanglement (Dong Yang)
 - Entanglement can Completely Defeat Quantum Noise (Jianxin Chen)



- Qubit Mapping and Routing
 - An Efficient Methodology for Mapping Quantum Circuits to the IBM QX Architectures ([Alwin Zulehner](#))
 - Tackling the Qubit Mapping Problem for NISQ-Era Quantum Devices ([Gushu Li](#))
 - LightSABRE: A Lightweight and Enhanced SABRE Algorithm (Henry Zou)
- Circuit transpilation and optimization
 - Automated Optimization of Large Quantum Circuits with Continuous Parameters (Yunseong Nam)
 - Reducing T-count with the ZX-calculus (Aleks Kissinger)
 - Quartz: Superoptimization of Quantum Circuits (Mingkuan Xu)



- Quantum Error Correction and Fault-tolerance

(Bivariate Bicycle Code Towards Experimental Realization)

- High-threshold and low-overhead fault-tolerant quantum memory (Sergey Bravyi)
- Demonstration of low-overhead quantum error correction codes (Ke Wang)

(Surface Code with Defects)

- Surface Code Error Correction on a Defective Lattice (Shota Nagayama)
- Low-Overhead Defect-Adaptive Surface Code with Bandage-Like Super-Stabilizers (Zuolin Wei)

(Surface Code and Lattice Surgery)

- A Game of Surface Codes: Large-Scale Quantum Computing with Lattice Surgery (Daniel Litinski)

允许合理使用 **AI** 工具辅助梳理思路、查阅资料、整理内容、辅助编程，**禁止直接抄袭照搬 AI 答案**。所有作业必须经过个人修改整合，加入自主思考，独立完成核心作答。